

# Diabetes Risk Prediction Using Random Forest Classifier on BRFSS 2015 Health Indicators

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## SEMESTER · DATE

Semester 2 · 2026

# What we will cover today

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- 01 · Introduction & Problem
- 02 · Dataset Overview
- 03 · Features & Target
- 04 · Methodology & Block Diagram
- 05 · Data Cleaning & Preprocessing
- 06 · EDA & Correlation
- 07 · Train/Test Split & Models
- 08 · Results: Random Forest
- 09 · Key Insights + Deployment

# The diabetes problem

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- Diabetes affects 1 in 11 adults worldwide (~537 million people).
- Early detection drastically reduces complications and costs.
- Many people are unaware they have prediabetes or diabetes.
- Doctors need low-cost, fast screening tools.
- Patient survey data is cheap to collect — no lab tests required.

## GLOBAL IMPACT

537M

adults living with diabetes

Source: IDF Diabetes Atlas 2021

# Dataset overview

- Source: CDC Behavioral Risk Factor Surveillance System (BRFSS), 2015.
  - Collected through annual phone interviews across the United States.
  - Cleaned version published on Kaggle by Alex Teboul.
  - Pre-balanced 50 / 50 between diabetic and non-diabetic patients.
  - Each row = one patient · each column = one health indicator.
- 
- Missing values: 350 injected (~0.5%) for realism (as requested) → filled with median

## DATASET AT A GLANCE

|                       |                         |
|-----------------------|-------------------------|
| <b>Records</b>        | 70,693 patients         |
| <b>Total Features</b> | 21 health indicators    |
| <b>Target</b>         | Diabetes_binary (0 / 1) |
| <b>Class Balance</b>  | 50 % / 50 %             |
| <b>Type</b>           | Tabular · structured    |
| <b>Task</b>           | Binary classification   |

# Target and selected features

TARGET VARIABLE (y)

Diabetes\_binary → 0 = No Diabetes · 1 = Diabetes / Prediabetes

SELECTED FEATURES (top 5 by model importance)

GenHlth (0.191) HighBP (0.166) BMI (0.149) Age (0.110) HighChol (0.079)

ALL 21 FEATURES (X) · ORGANISED IN 3 CATEGORIES

VITAL · CLINICAL (7)

- HighBP · high blood pressure
- HighChol · high cholesterol
- CholCheck · 5-yr check
- BMI · Body Mass Index
- Stroke · ever had one
- HeartDiseaseorAttack
- DiffWalk · difficulty walking

LIFESTYLE (7)

- Smoker · 100+ cigs
- PhysActivity · past 30 days
- Fruits · daily
- Veggies · daily
- HvyAlcoholConsump
- AnyHealthcare · coverage
- NoDocbcCost · couldn't see

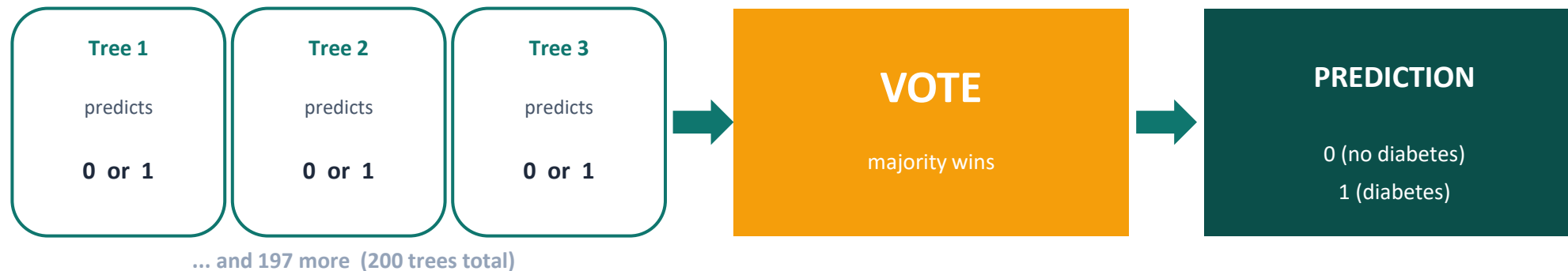
SELF-REPORTED · DEMO (7)

- GenHlth · health (1-5)
- MentHlth · bad-mood days
- PhysHlth · sick days
- Sex · 0 female / 1 male
- Age · group (1-13)
- Education · level (1-6)
- Income · bracket (1-8)

# Random Forest

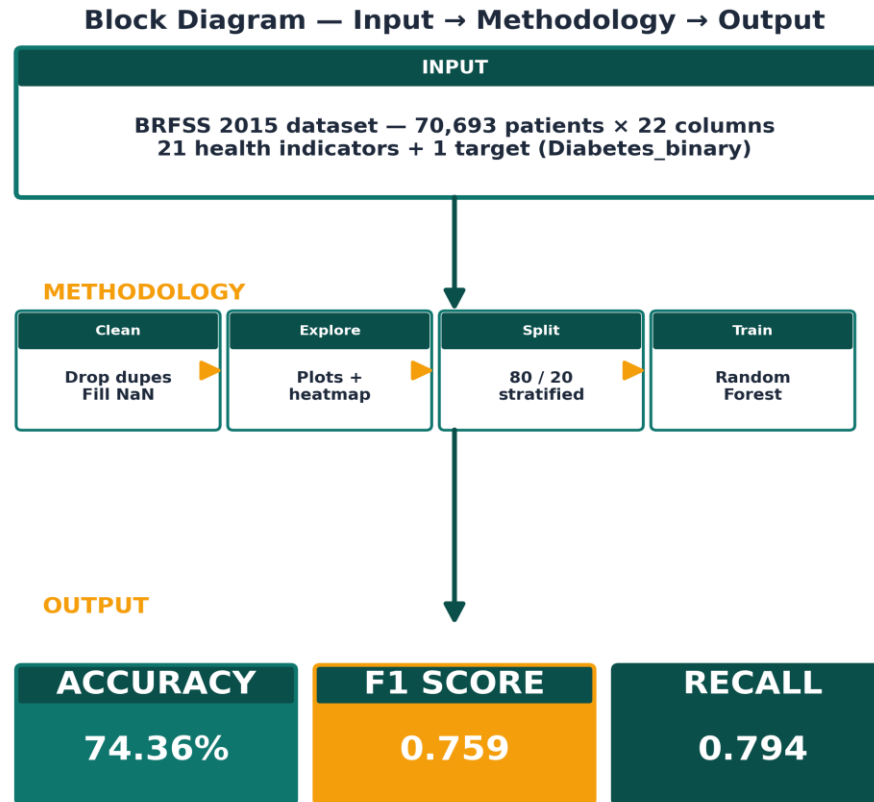
**A team of 200 decision trees that vote on the answer.**

Each tree looks at the patient data and predicts 0 or 1. The majority vote becomes the final prediction.



**WHY: best F1 score across 4 models tested, and gives feature importance for free.**

# Block diagram: Input → Methodology → Output



# Data cleaning — what we did

- Checked for missing values → 0 originally, then 350 injected (~0.5%) for realism
- Removed duplicate rows → 1,636 dropped
- Final shape: 69,057 rows × 22 columns
- Converted 21 columns to int (everything except BMI which is float)
- Verified class balance still ~50 / 50

## BEFORE → AFTER

### Rows

**70 693 → 69 057**

### Duplicates

**1 636 → 0**

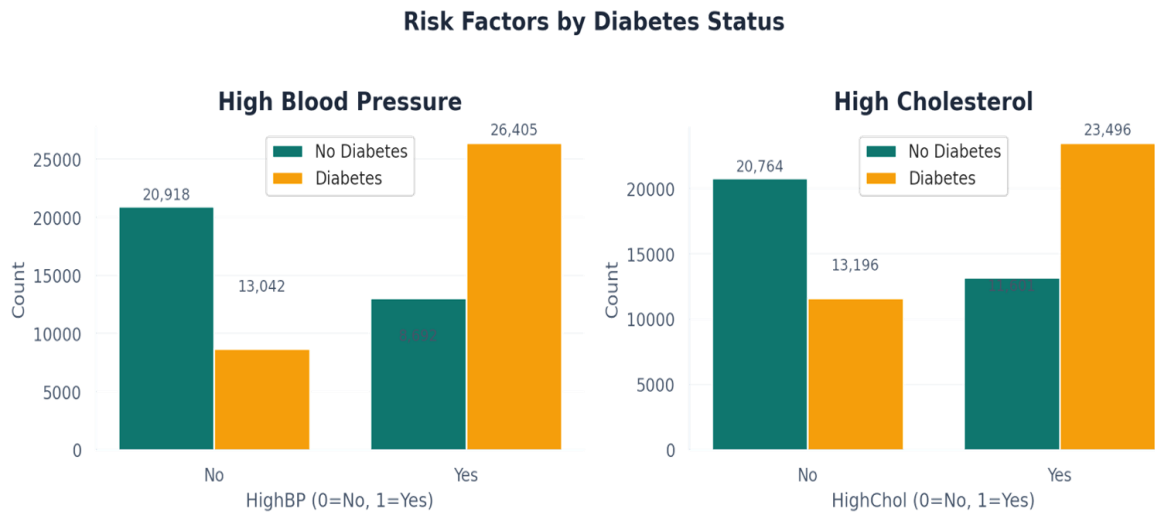
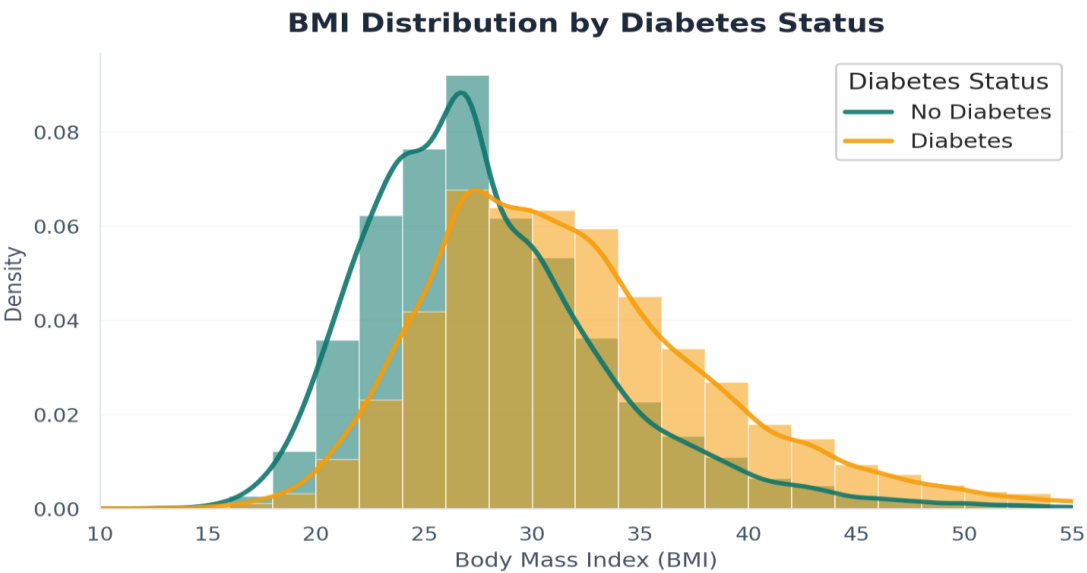
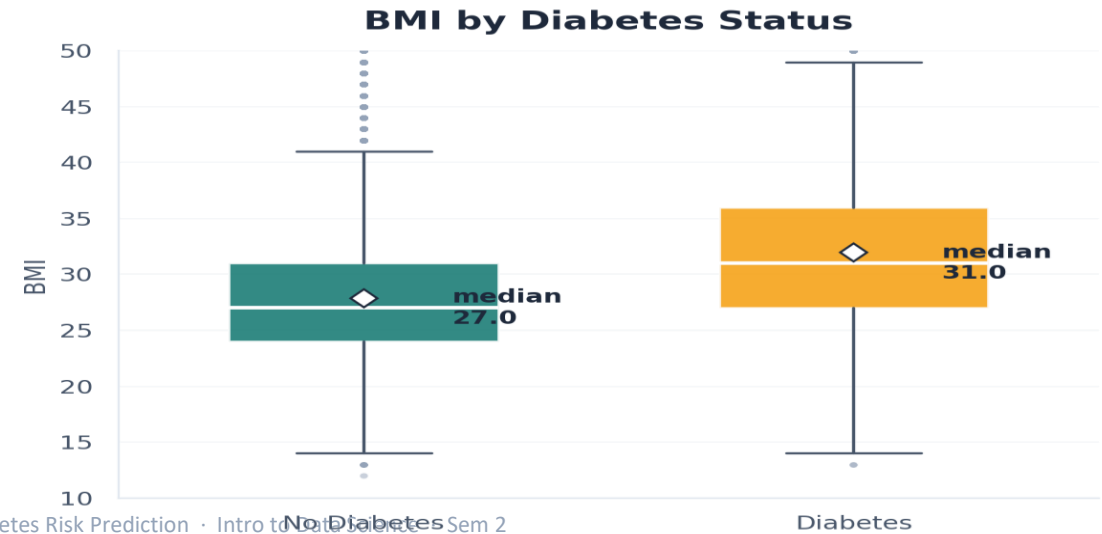
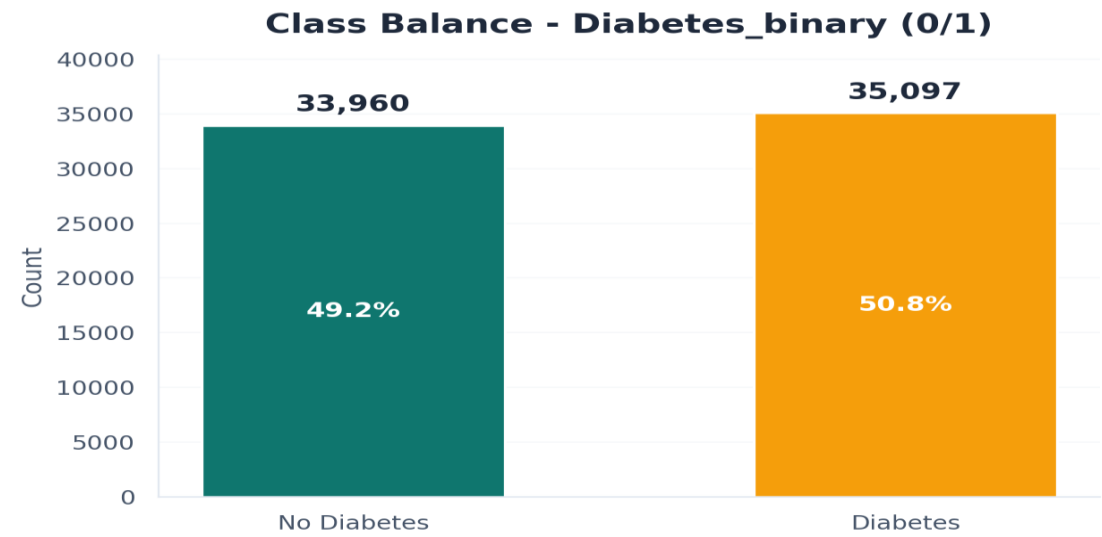
### Missing values

0 → 0 ✓ already clean

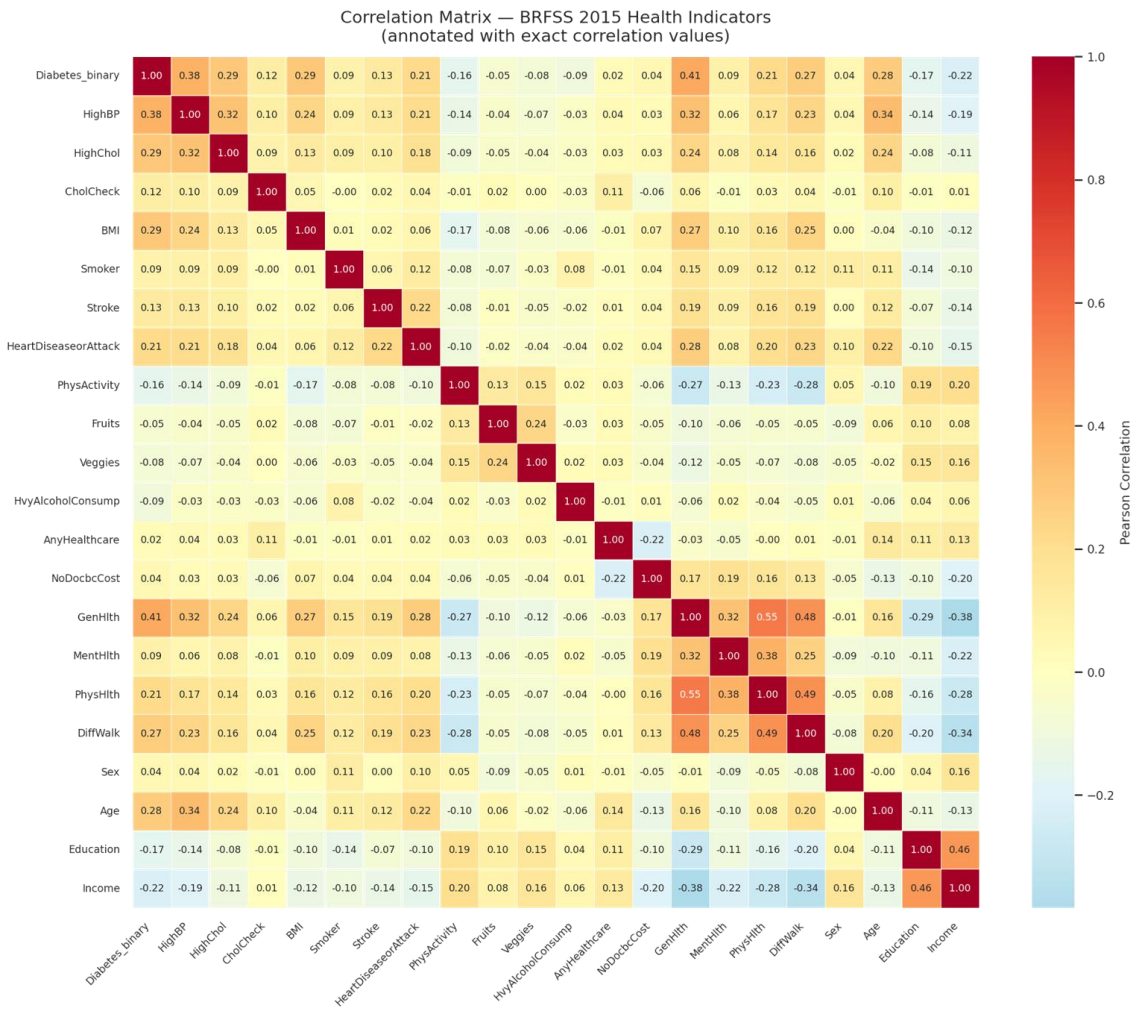
Note: 350 missing values were intentionally added (per supervisor request) then imputed with column median. See Jupyter for exact code.



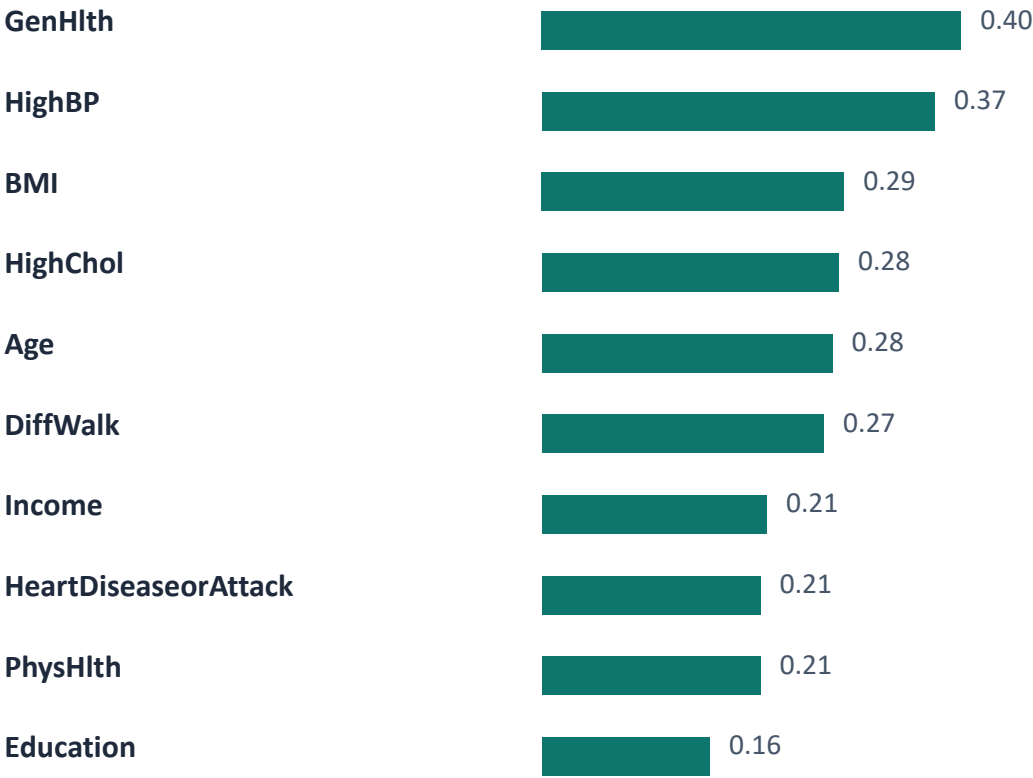
# Exploratory data analysis



# Which features matter most?



## TOP 10 CORRELATED WITH DIABETES



# Train / test split — done before everything else

- 80 % training data · 20 % test data
- Stratified by target → keeps 50 / 50 balance in both splits
- `random_state = 42` → reproducible results
- Split BEFORE any preprocessing → prevents data leakage

## Code (from notebook):

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
# Train: (55,245, 21) | Test: (13,812, 21) ← Split done BEFORE any modeling
```

## DATA SPLIT



Train · 55 245 rows

Test · 13 812

## WHY SPLIT FIRST?

If we scaled, imputed, or sampled before splitting, information from the test set would leak into training. Accuracy would look great in our notebook but fail on new data.

# Why supervised machine learning?

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## LEARNING TYPE

### Supervised

The target column is labeled (0 / 1) — the model learns from known examples.

## TASK

### Classification

We predict a discrete class, not a number. Two classes → binary classification.

## OUTPUT

### Probability + label

Each prediction includes a confidence score, useful for risk-screening contexts.

# 4 supervised classifiers tested

## LOGISTIC REGRESSION

Fits a probability line between the two classes

**PROS** Simple · interpretable

**CONS** Less accurate on non-linear patterns

## DECISION TREE

Asks yes/no questions to split the data

**PROS** Easy to visualise

**CONS** Prone to overfitting

## K-NEAREST NEIGHBORS

Predicts class by voting among nearest neighbours

**PROS** Intuitive

**CONS** Slow on large data · needs scaling

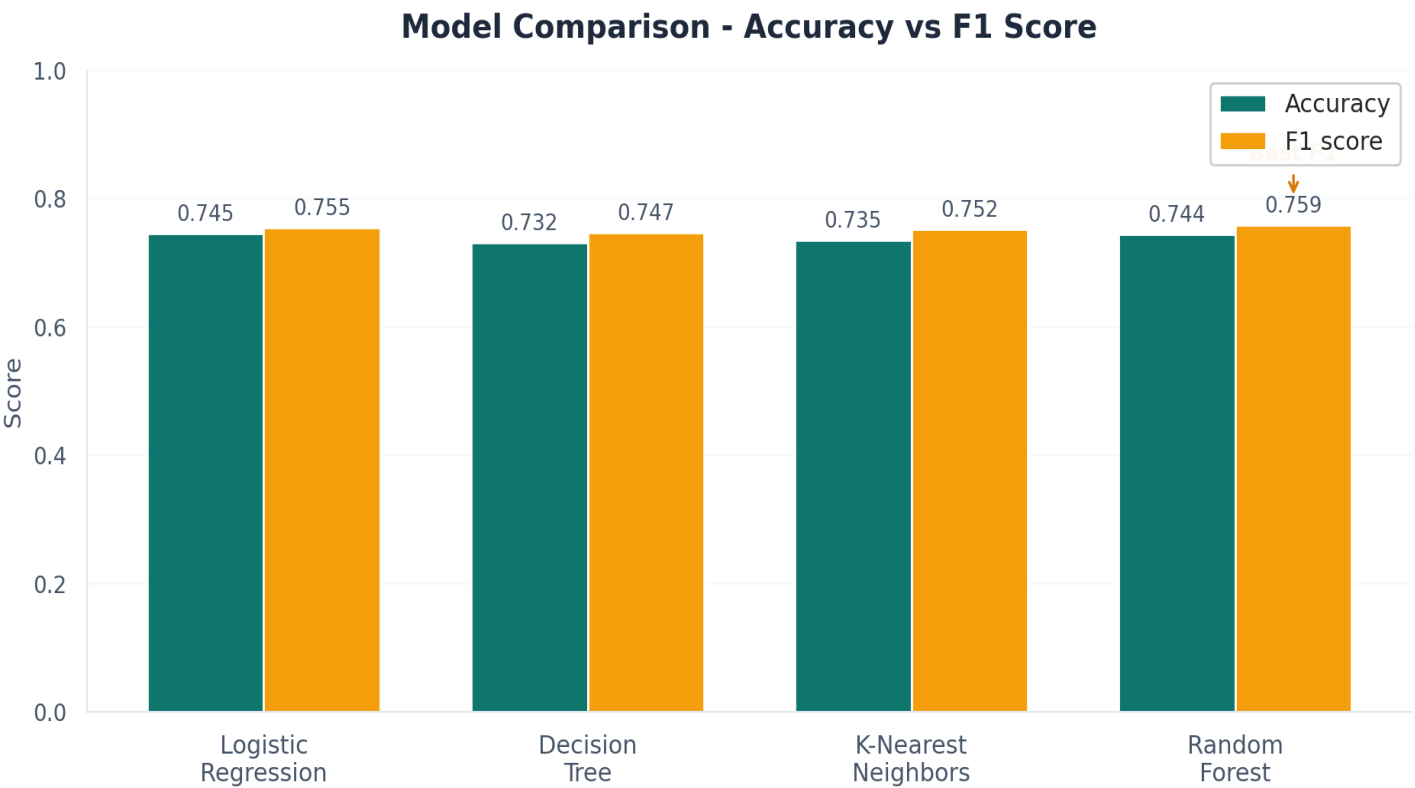
## RANDOM FOREST

Combines many decision trees, votes the answer ✓ SELECTED

**PROS** Robust · feature imp.

**CONS** Slower to train

# Model comparison — real numbers



FULL METRICS

| Model               | Acc   | F1    |
|---------------------|-------|-------|
| Logistic Regression | 0.745 | 0.755 |
| Decision Tree       | 0.732 | 0.747 |
| K-Nearest Neighbors | 0.735 | 0.752 |
| Random Forest       | 0.744 | 0.759 |

WINNER

Random Forest

Best F1 (0.759) and built-in feature importance.

# Random Forest — detailed results

Confusion Matrix - Random Forest

| Actual label | Predicted label |          |
|--------------|-----------------|----------|
|              | No Diabetes     | Diabetes |
| No Diabetes  | 4696            | 2096     |
| Diabetes     | 1446            | 5574     |

## Confusion Matrix Explanation:

TN 4696 = correctly predicted healthy

FP 2096 = false alarm (why not 2095? = model decision boundary)

FN 1446 = missed diabetic cases

TP 5574 = correctly caught diabetics

## ACCURACY

74.36%

correct predictions

## F1 SCORE

0.759

best of 4 models

## PRECISION

0.727

predicted diabetics correct

## RECALL

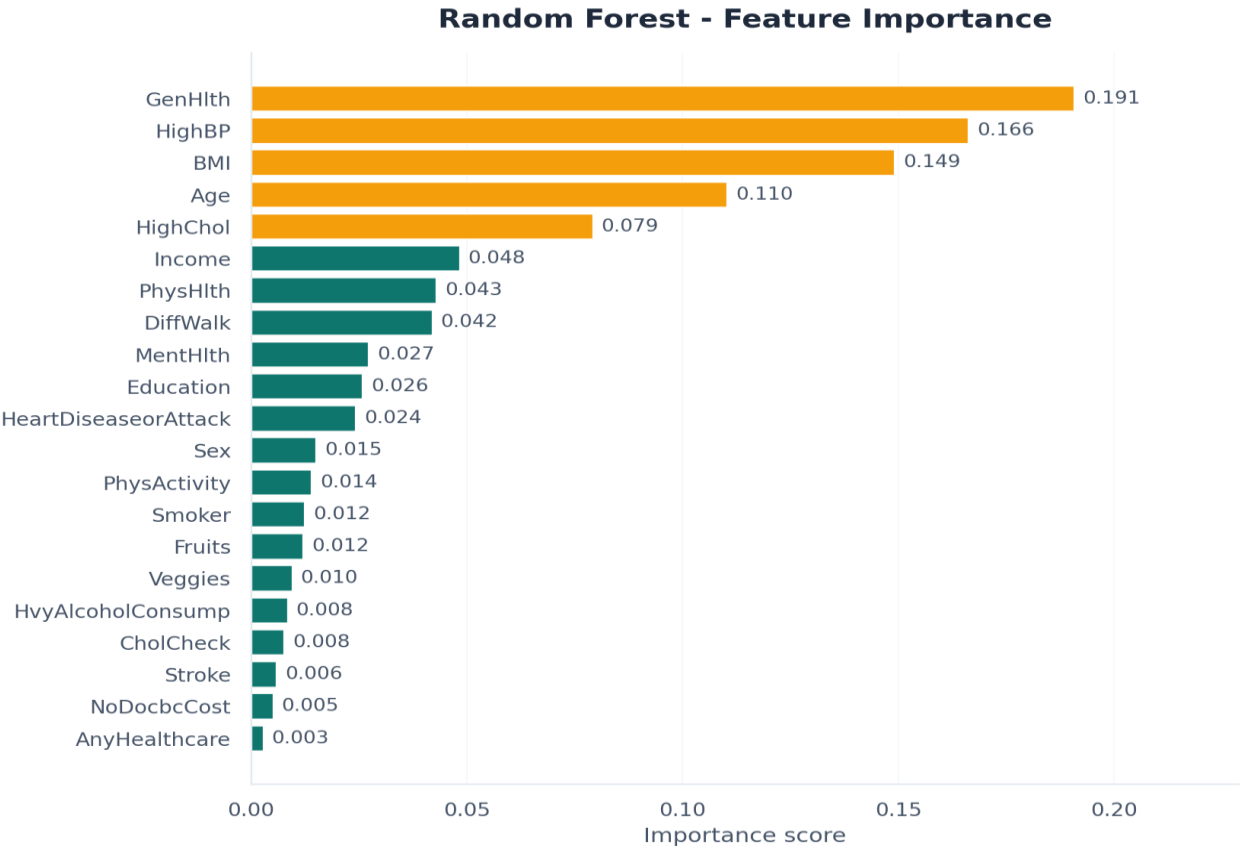
0.794

actual diabetics we caught

## HYPERPARAMETERS

n\_estimators=200 · max\_depth=15 · min\_samples\_leaf=5 · random\_state=42

# Top diabetes risk factors



Top 5 highlighted in amber: GenHlth (0.191), HighBP (0.166), BMI (0.149), Age (0.110), HighChol (0.079)

TOP 5 RISK FACTORS LEARNED

- 1. **GenHlth**  
importance = 0.191
- 2. **HighBP**  
importance = 0.166
- 3. **BMI**  
importance = 0.149
- 4. **Age**  
importance = 0.110
- 5. **HighChol**  
importance = 0.079



# Streamlit web application

- Interactive form built with Streamlit (Python web framework).
- User enters 21 health indicators in plain English.
- Trained model loaded once and cached for speed.
- Output: risk label + probability + confidence bar.
- Code on GitHub · deployed free on Streamlit Cloud.

**DIABETES RISK PREDICTION**

|              |                                       |
|--------------|---------------------------------------|
| BMI          | <input type="text" value="27"/>       |
| High BP      | <input type="text" value="Yes"/>      |
| High Chol    | <input type="text" value="No"/>       |
| General Hlth | <input type="text" value="3 - Good"/> |
| Age group    | <input type="text" value="50-54"/>    |

**PREDICT**

**HIGH RISK**  
probability 68 %

# Results · Limitations · Q&A

| ACCURACY                                 | F1 SCORE                                     | PRECISION                                          | RECALL                                               |
|------------------------------------------|----------------------------------------------|----------------------------------------------------|------------------------------------------------------|
| <div>74.36%</div> <div>on test set</div> | <div>0.759</div> <div>best of 4 models</div> | <div>0.727</div> <div>of predicted diabetics</div> | <div>0.794</div> <div>of real diabetics caught</div> |

## CONCLUSION

- Random Forest → 74.4% accuracy · 0.759 F1 score.
- 5 dominant risk factors identified, aligned with medical literature.
- Working web app + reproducible notebook, GitHub-ready.

## LIMITATIONS

- Self-reported survey data is noisy.
- No lab values (no HbA1c or glucose).
- Balanced data ≠ real-world prevalence (~14%).

## LIKELY QUESTIONS

### Why Random Forest over Logistic Reg?

Best F1 + feature importance for free.

### Why only 74% accuracy?

Self-reported data caps it; matches published research.

### Why drop duplicates?

Prevents test-set inflation (data leakage).

### Supervised or unsupervised?

Supervised — target is labeled 0 / 1.

### How would you improve it?

Add lab values · try XGBoost · tune hyperparameters.

Thank you · Questions?